

PU-I Biology — Chapter 3: Plant Kingdom

Worksheet: Classes of Algae

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Name: _____

Section: _____

Date: _____

Section A — Fill in the Blanks

5 Questions · Classes of Algae

1. Chlorophyceae members are commonly called _____ algae.
2. Phaeophyceae members are commonly called _____ algae.
3. Rhodophyceae members are commonly called _____ algae.
4. Brown algae store food as laminarin or _____.
5. Red algae store food as _____ starch.

Section B — One-word Answers

1. Name the pigment responsible for the red colour in Rhodophyceae. _____
2. What are the storage bodies containing protein and starch in Chlorophyceae chloroplasts?

3. Name the substance coating the brown algal cell wall. _____
4. Name the leaf-like photosynthetic organ of a brown alga. _____
5. Give one example each of Chlorophyceae and Rhodophyceae. _____

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Section C — Match the Following

Column A	Column B
1. Chlamydomonas	a. Chlorophyceae
2. Sargassum	b. Phaeophyceae
3. Porphyra	c. Rhodophyceae
4. Chara	d. Phaeophyceae
5. Fucus	e. Chlorophyceae

Section D — True / False

1. Chlorophyceae contains chlorophyll a and b. _____
2. Phaeophyceae is mainly a freshwater group. _____
3. Rhodophyceae reproduces sexually by oogamy. _____
4. The frond is the attachment structure of a brown alga. _____
5. Pyrenoids are found in most Chlorophyceae chloroplasts. _____

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Section E — Short Answer

1. Differentiate between the pigments of Chlorophyceae, Phaeophyceae, and Rhodophyceae.

2. Describe the plant body of a typical brown alga.

3. What is the significance of pyrenoids?

Section F — HOTS (Higher Order Thinking)

1. Explain why red algae can survive in deeper marine water than green algae, using their pigment composition.

2. A marine alga up to 90 m long has a holdfast, stipe, and frond, and stores food as mannitol. Identify its class and justify your answer.

Teacher's Signature: _____ Date: _____

Teacher's Remarks:
